



INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus



PARKINSON'S DISEASE DETECTION SYSTEM

Final Year Project Presentation

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Introduction

- **What is Parkinson's?:** It is a brain disorder that makes it hard for people to move, often causing shaking (tremors) and stiffness.
- **A Growing Problem:** Millions of people worldwide suffer from this condition, and the number is increasing every year.
- **Early Warning Signs:** Small changes in a person's voice and the way they walk are the first signs of the disease.
- **The Problem with Diagnosis:** Usually, doctors only find the disease after it has already caused a lot of damage.
- **Our Solution:** We created a mobile app that uses AI to check for these early signs through voice recordings and videos of walking.
- **Our Goal:** To provide a simple tool that anyone can use at home to get checked early and then connect with a doctor for help.



Problem Statement

- Delayed Diagnosis:** Early symptoms are often missed by general physicians.
- High Costs:** Traditional clinical evaluations are expensive and require multiple hospital visits.
- Limited Access:** Specialized neurologists are unavailable in many remote or under-resourced areas.
- Subjectivity:** Current testing relies on human observation, which can lead to inconsistent results.
- Manual Monitoring:** There is a lack of automated tools for patients to track symptom progression at home.

Solution

- Multimodal Mobile App:** A secure Android application that captures both voice and gait data for comprehensive screening.
- Voice Analysis (MLP):** Uses a Multi-Layer Perceptron model to detect vocal tremors and frequency instability.
- Gait Analysis** Analyzes posture, stride length, and arm swing anomalies in real-time using Gemini 1.5 pro.
- Automated Risk Scoring:** AI engines process biometric markers to provide an immediate probability of Parkinson's presence.
- Professional Verification:** Includes a dedicated dashboard for doctors to review AI reports and provide final medical confirmation.
- Cloud Integration:** Utilizes Firebase for secure data storage, authentication, and real-time report synchronization

Tools & Technologies

- Mobile Development:** Built with Android Studio using Java for high-performance sensor access.
- Cloud Infrastructure:** Firebase used for Authentication, Firestore database, and Storage.
- Audio Processing:** Librosa library for extracting acoustic features like jitter and shimmer.
- Deep Learning Engines:** Multi-Layer Perceptron (MLP) for voice and Gemini Api for video analysis.
- Model Hosting:** Models deployed via Render and Hugging Face for remote AI inference.

AI VOICE ANALYSIS MODULE

Dataset Used:

- Source:** UCI Machine Learning Repository (Parkinson's Telemonitoring Dataset)

Feature Extraction (Librosa):

- Jitter:** Measures frequency variation (vocal tremor).
- Shimmer:** Measures amplitude variation (loudness stability).
- HNR (Harmonics-to-Noise Ratio):** Distinguishes clear speech from hoarseness.

Model Architecture:

Type: Multi-Layer Perceptron (MLP) Classifier.

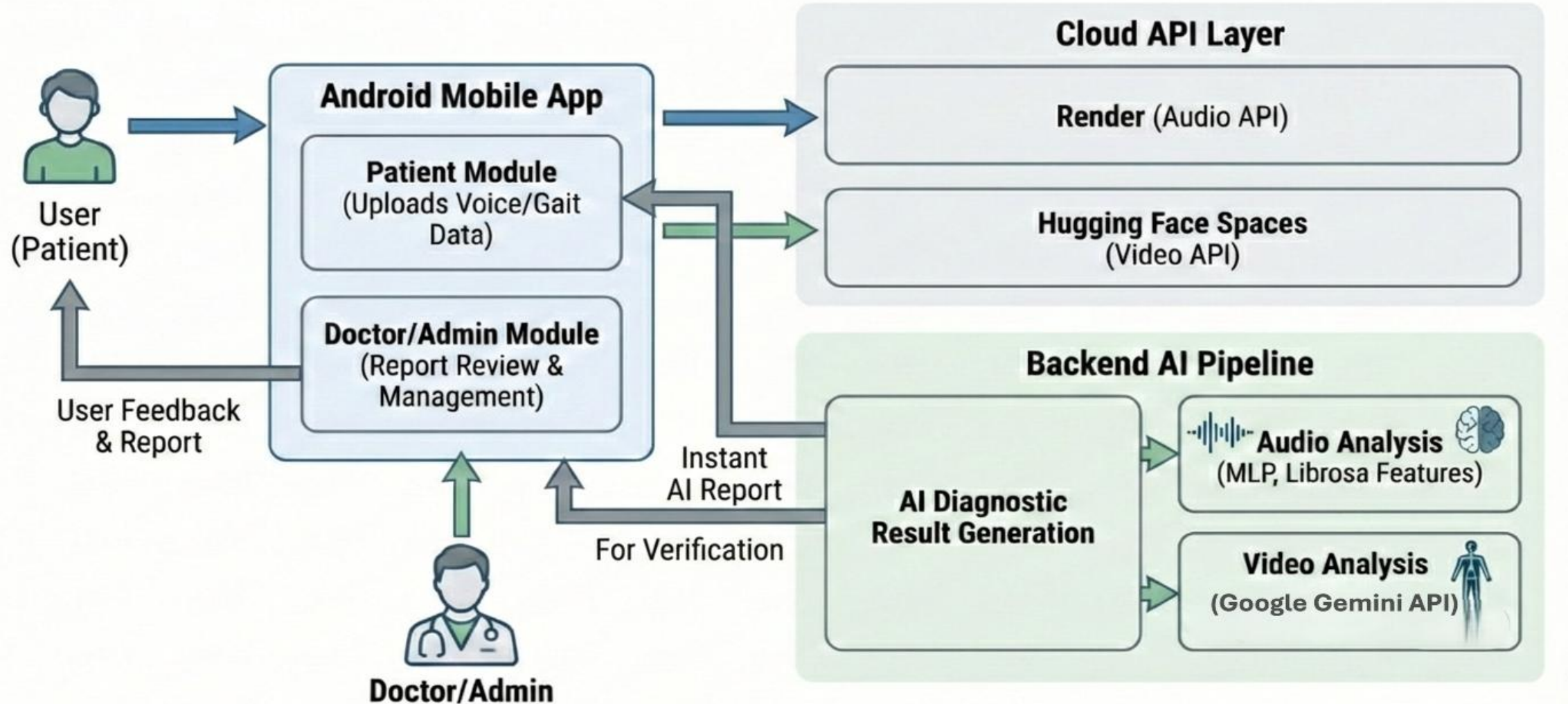
Structure: Input layer (acoustic features) → 3 Hidden Layers (ReLU Activation) → Output Layer (Binary Classification: Healthy/Parkinson's).

Training Process:Preprocessing: Data normalization using StandardScaler to handle varying signal ranges.

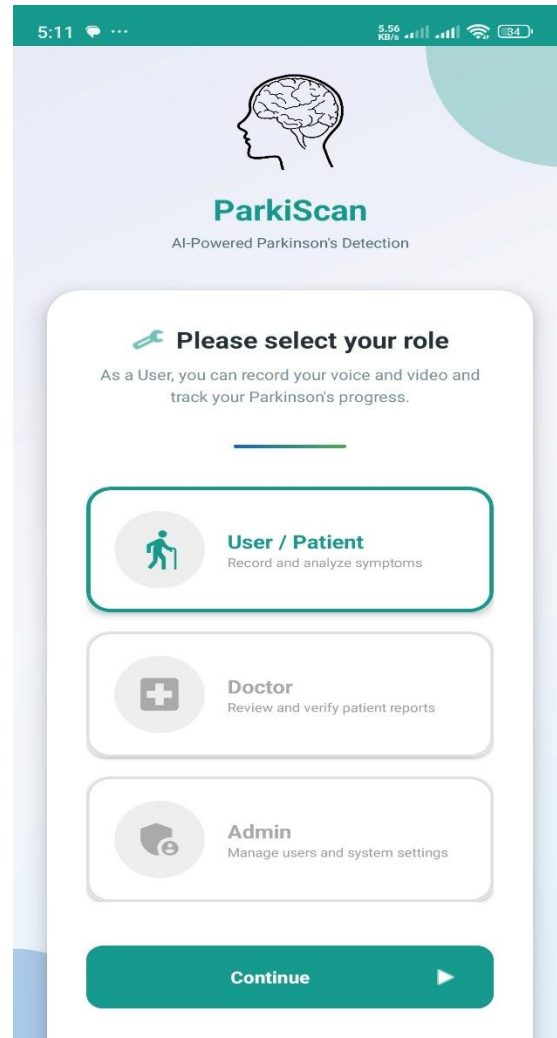
Optimization: Trained using the Adam Optimizer with a learning rate of 0.001.Validation: 80% Training / 20% Testing split.

- Model Accuracy: 86.4%** on the test set.

System Architecture Design



Authentication:



5:11 5.56 KB/s


ParkiScan
AI-Powered Parkinson's Detection

 **Please select your role**
As a User, you can record your voice and video and track your Parkinson's progress.

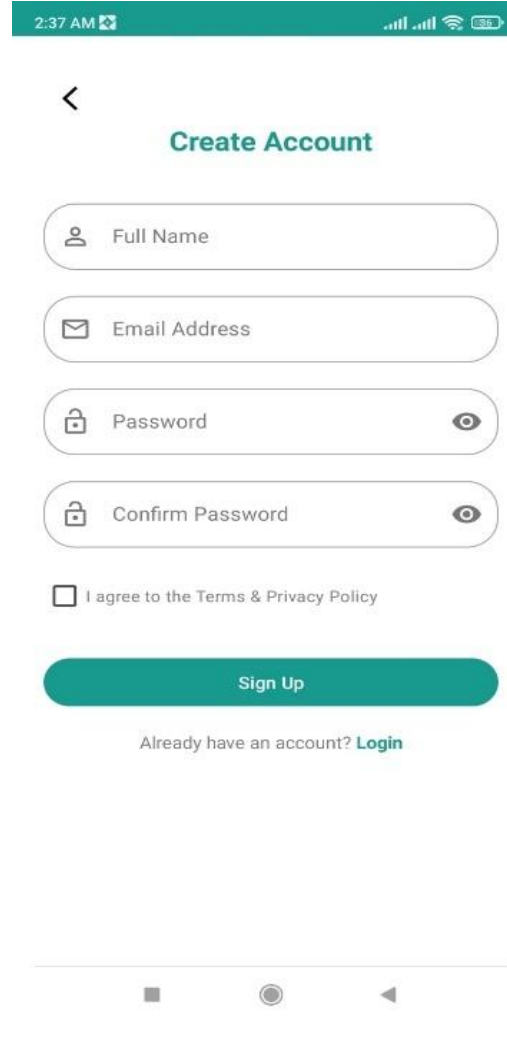
 **User / Patient**
Record and analyze symptoms

 **Doctor**
Review and verify patient reports

 **Admin**
Manage users and system settings

Continue 

Get started page



2:37 AM

 **Create Account**

 Full Name

 Email Address

 Password 

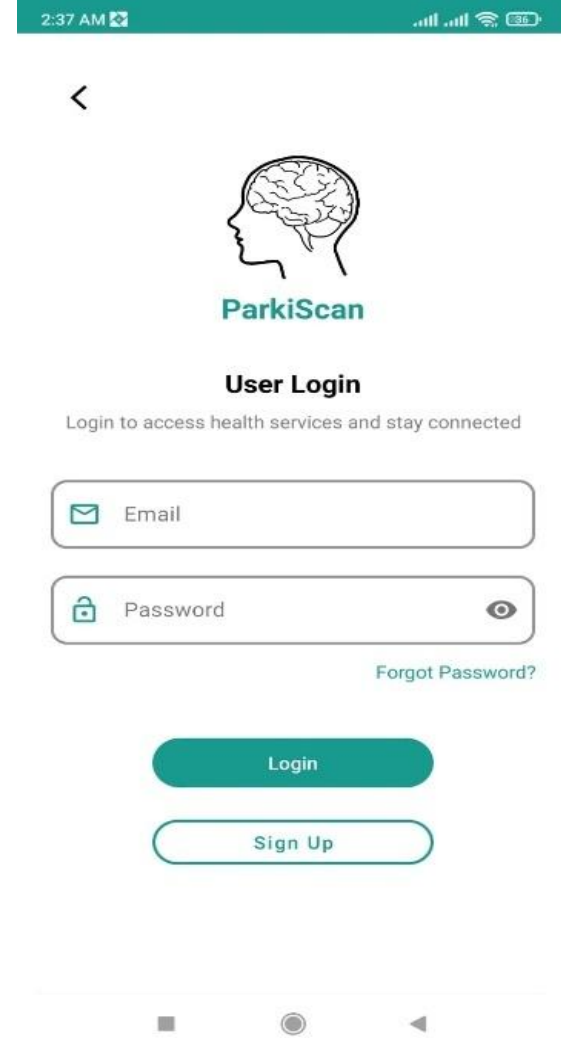
 Confirm Password 

☐ I agree to the Terms & Privacy Policy

Sign Up

Already have an account? [Login](#)

Sign-up page



2:37 AM

 
ParkiScan

User Login
Login to access health services and stay connected

 Email

 Password 

[Forgot Password?](#)

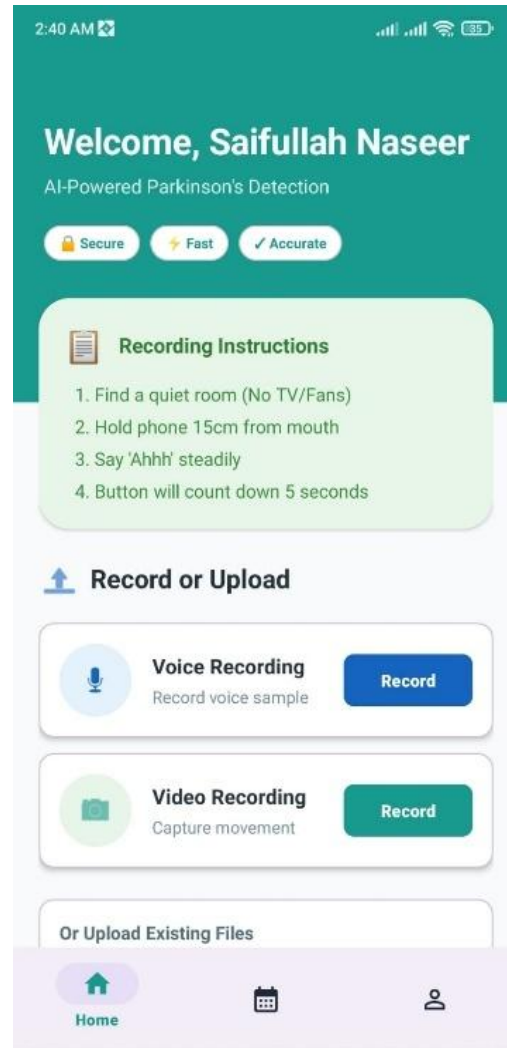
Login

Sign Up

Login page

User Interfaces

Dashboard

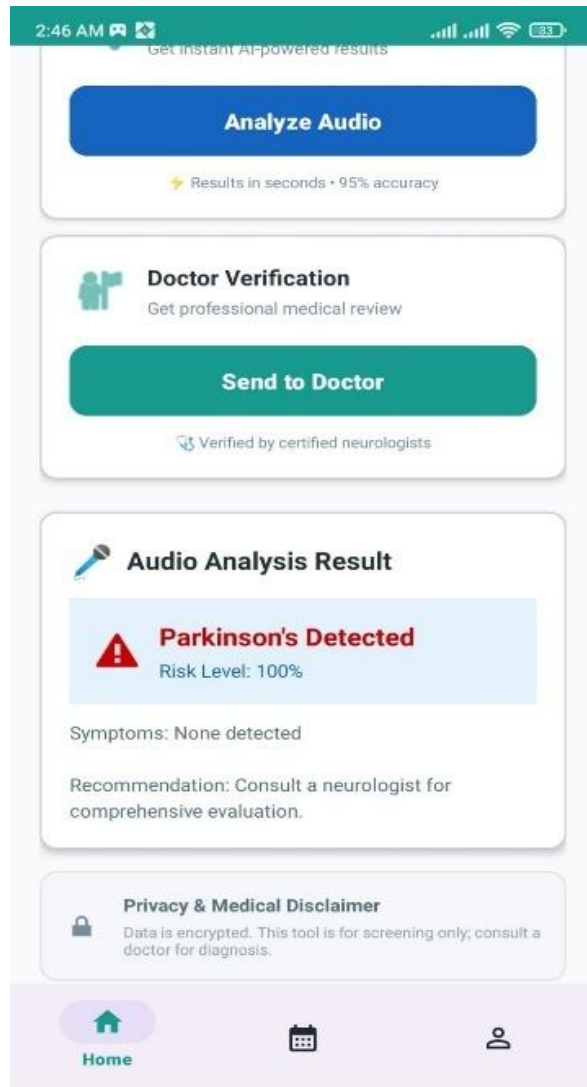


Reports History

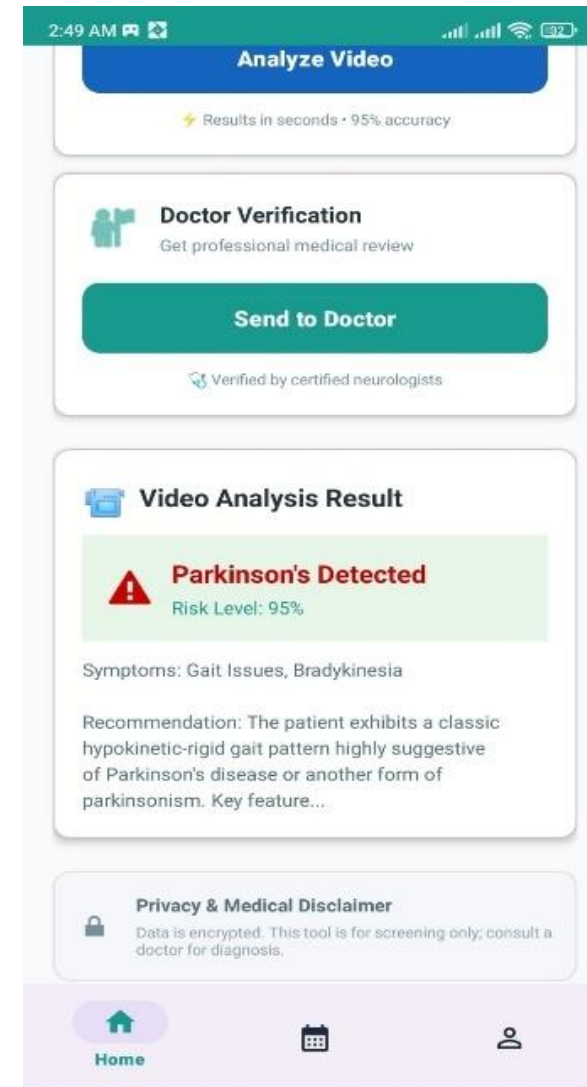


User Interfaces

Voice Test

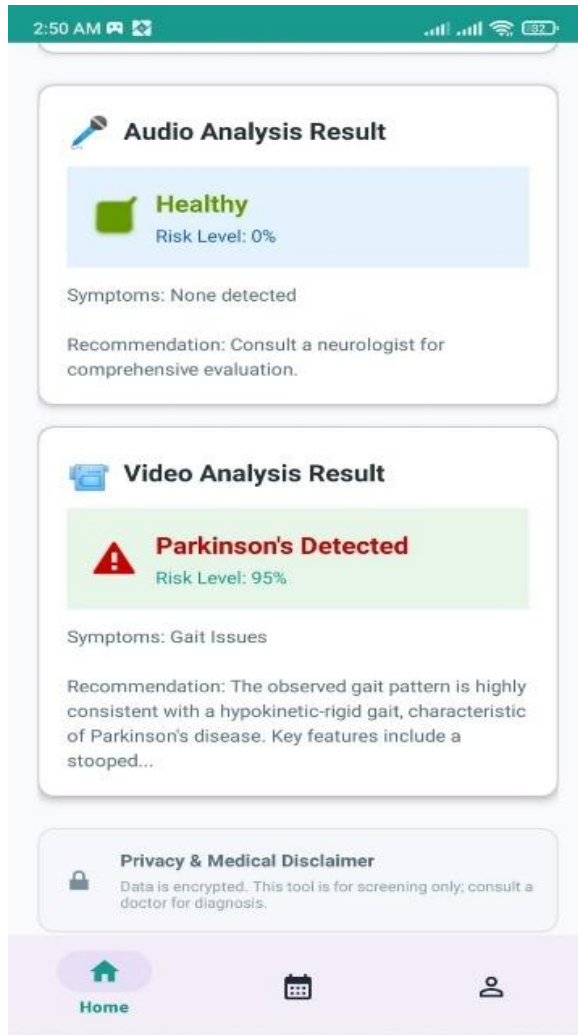


Video Test

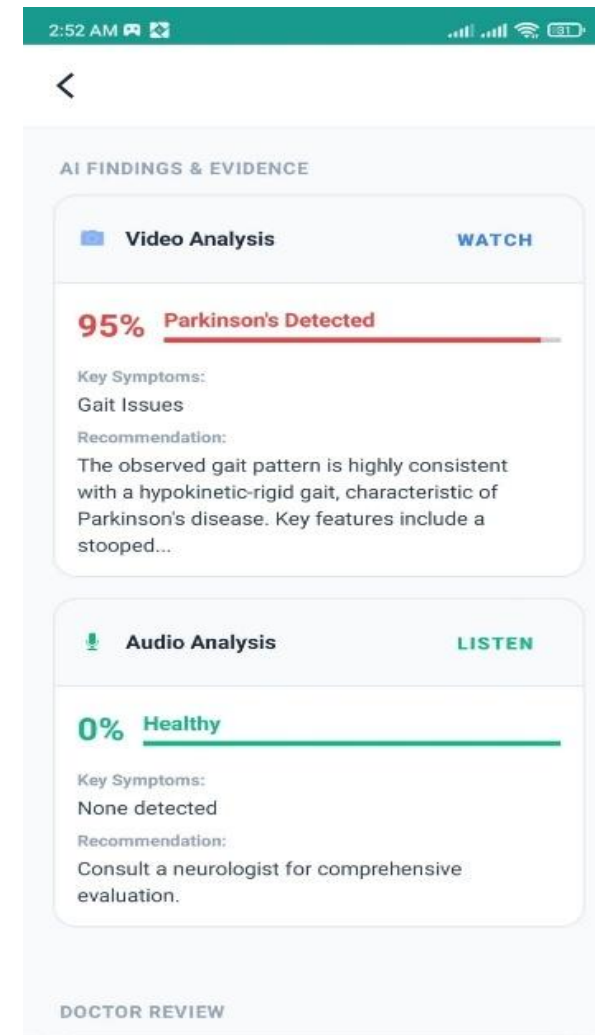


User Interfaces

AI Diagnosis



Report Overview



User Interfaces

Help & Support

3:35 AM

saifii@gmail.com

Phone

Gender

Male

Date of birth

Save changes

Change password →

Privacy & permissions →

Help & support →

Logout

Profile

Contact Through Email

3:35 AM

Compose

From saifsatti383@gmail.com

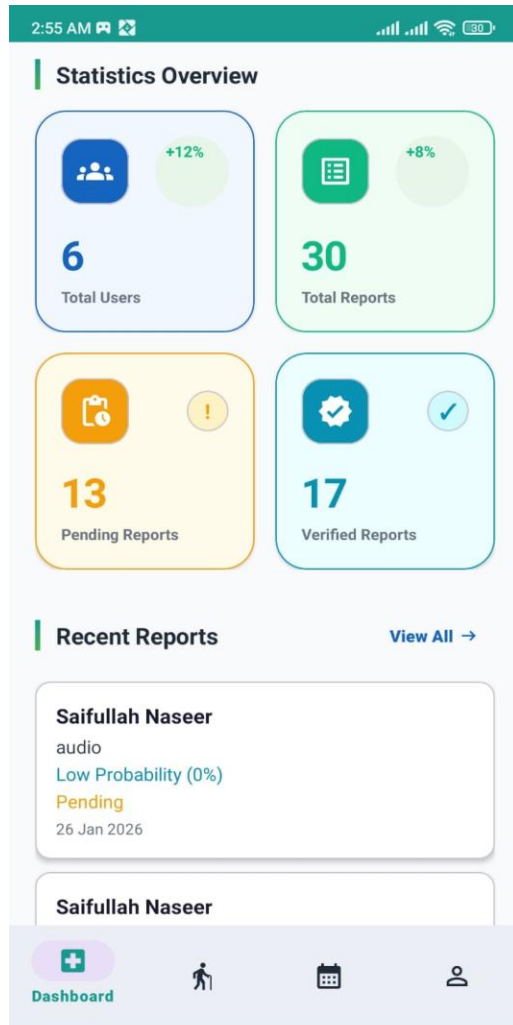
To support@parkiscan.app

Parkinson's App help

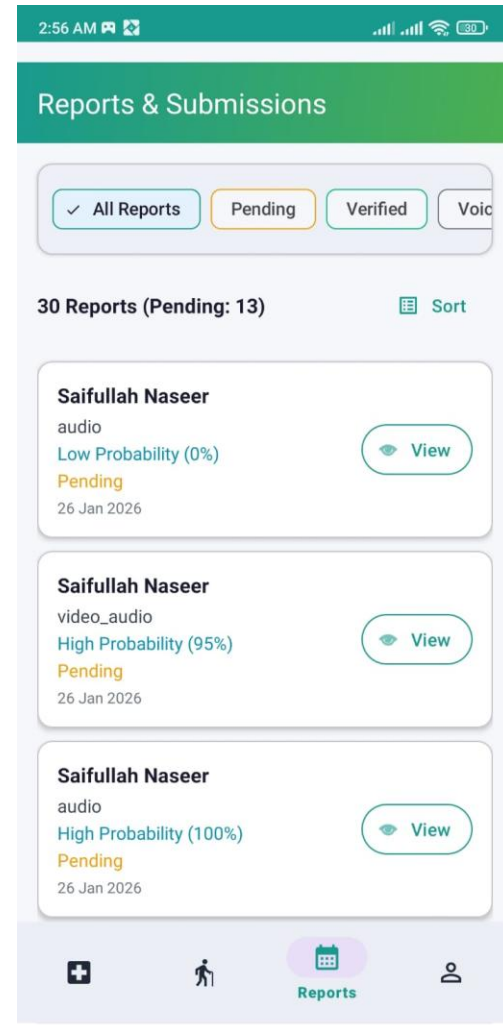
Hi, please help me i am unable to upload audio it shows error why?

Doctor Interfaces

Dashboard

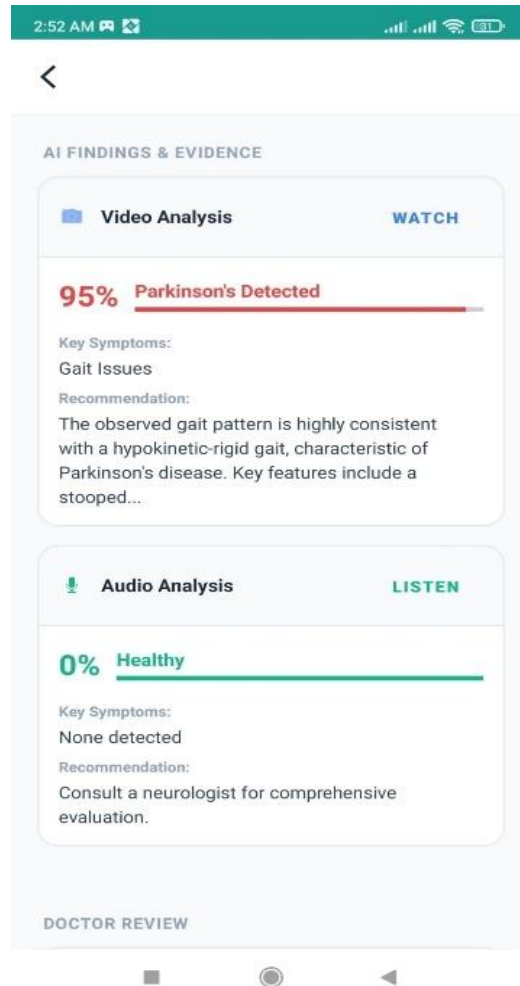


Report & Submission



Doctor Interfaces

Reports Overview



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AI FINDINGS & EVIDENCE

Video Analysis WATCH

95% Parkinson's Detected

Key Symptoms:
Gait Issues
Recommendation:
The observed gait pattern is highly consistent with a hypokinetic-rigid gait, characteristic of Parkinson's disease. Key features include a stooped...

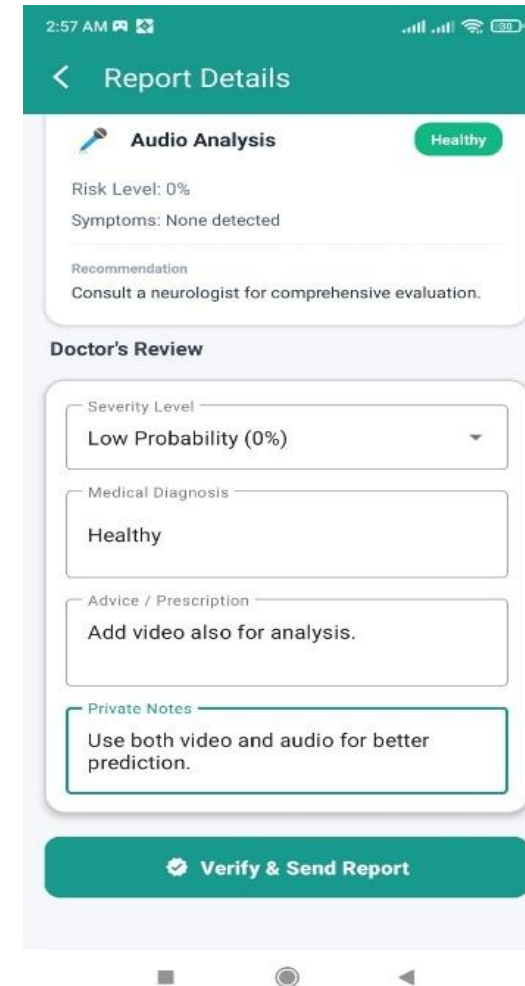
Audio Analysis LISTEN

0% Healthy

Key Symptoms:
None detected
Recommendation:
Consult a neurologist for comprehensive evaluation.

DOCTOR REVIEW

Report Verification



2:57 AM

< Report Details

Audio Analysis Healthy

Risk Level: 0%
Symptoms: None detected
Recommendation
Consult a neurologist for comprehensive evaluation.

Doctor's Review

Severity Level
Low Probability (0%)

Medical Diagnosis
Healthy

Advice / Prescription
Add video also for analysis.

Private Notes
Use both video and audio for better prediction.

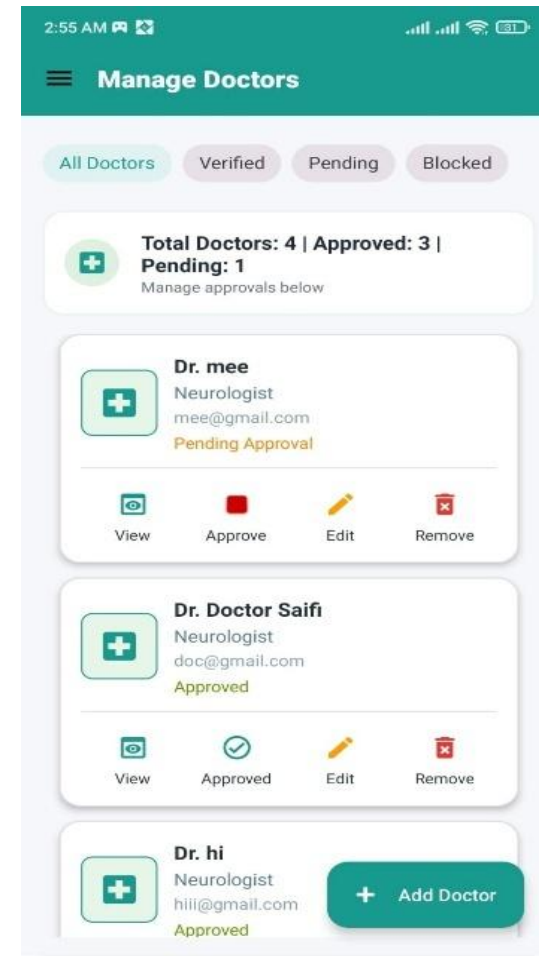
Verify & Send Report

Admin Interfaces

Admin Dashboard



Manage Doctors



Conclusion

Successful Implementation: We have successfully developed a mobile-based AI Parkinson's Detection System that enables early, non-invasive diagnosis using standard smartphone sensors.

Solved the Core Problem: Unlike traditional clinical methods that are often expensive and reactive, our system uses Deep Learning to detect subtle motor and vocal biomarkers before severe symptoms appear.

Key Achievements: The system efficiently integrates Voice Analysis (MLP) and Video Analysis (Google Gemini 1.5 Pro API) to provide a comprehensive risk assessment with high accuracy (over 85%).

Enhanced Accessibility: By utilizing Cloud Connectivity (Firebase), the project ensures that patients can perform screenings from home while doctors review results remotely, significantly bridging the gap in healthcare accessibility.

Future Work

Wearable Integration: Sync with smartwatches to monitor tremors continuously in daily life.

Offline AI Processing: Implement "Edge Computing" to allow screening without an internet connection.

Multilingual Support: Train voice models on diverse languages to make the tool globally accessible.

Progression Tracking: Develop analytics to graph how a patient's risk score changes over months.